Data types

L02
FACT FILE

Specification: students should be able to

- Identify the following data types: numeric (integer and real), date/time, character and string
- Explain why data must be defined as a particular data type before it is processed
- Select and justify the most appropriate data type for a given item of data

CCEA fact file: Digital Data 01: Representing Data

Key words: data type, numeric, integer, real, double, date/time, date format, character, string, character set, type check, currency, Boolean (Yes/No), leading zero

Field	Example value	Data type
Surname	Maguire	String
Age	15	Integer
Height (m)	1.62	Real
Form initial	B	Character
Date of birth	12/09/2010	Date/time
Phone	07700 900123	String

Phone numbers are strings: the leading 0 must be kept and no maths is done on them.

A pupil record with each field matched to its data type

What is a data type?

A data type tells the computer what kind of data an item is, for example a whole number, a date or some text. Data must be defined as a particular type before it is processed. This matters for three main reasons. First, it decides what processing is possible: numbers can be added and averaged, dates can be put in order or used to work out how many days are left, and text can be searched or joined together. Second, it helps the computer store data efficiently, because each type is stored in a particular way. Third, it supports validation. If a field is set to integer, a type check will reject an entry such as 'twelve', which helps stop errors getting into the system. Choosing the right type when a database, spreadsheet or program is set up saves problems later.

Numeric data: integer and real

Numeric data is data that you can do calculations with. The specification splits it into two types. An integer is a whole number with no decimal or fractional part. It can be positive or negative, so 28, 0 and -4 are all integers. Integers suit counts, such as the number of pupils in a class, tickets sold for a school show or a score of 51 in a test. A real number can have a decimal or fractional part, such as 1.62, 3.99 or -4.6. Real numbers can also be positive or negative, and the set of real numbers includes whole numbers too. Real is used for measurements and money, such as heights, temperatures and prices. In C#, the course programming language, int stores integers and double stores real numbers. In a database, money is often stored using a currency type, which is shown with a currency symbol and two decimal places.

Date/time, character and string

The date/time type stores dates, times or both. Because the computer knows the value is a date, it can sort records by date and calculate with them, such as finding everyone whose membership ends this month. Date/time also controls how the value is displayed. Software offers ready-made formats, for example a long date (Saturday, 12 September 2026), a medium date (12-Sep-2026) or a short date (12/09/2026), and similar long, medium and short time formats; users can also create their own. A character is one single letter, digit or symbol, such as a grade 'B' or 'Y' for yes. It is stored using its code from the character set, such as ASCII. A string is a sequence of characters, such as a name, address or postcode. Strings can vary in length. Any digits in a string are treated as text, so they cannot be used in calculations.

Choosing and justifying a data type

Many exam questions give you an item of data and ask for the most suitable type and a reason. Work through a few questions in order. Will calculations be done with it? If so it is numeric: real if it needs a decimal part, integer if not. Is it a date or time? Then use date/time. Is it always exactly one character? Then character. Everything else is a string. Watch for 'numbers' that are really text. A mobile phone number such as 07700 900123 must be a string, because storing it as an integer would remove the leading zero, the space would not be allowed and nobody ever adds phone numbers together. Postcodes such as BT92 1AB mix letters and digits, so they are strings too. A pupil's age is an integer, but their date of birth is date/time, because the age can be worked out from it and it stays correct as time passes.

T Key terms from the mark schemes

Data type
The kind of data an item is (such as integer, real, date/time, character or string), which decides how it is stored and processed.

Real
Data type for numbers that can have a decimal or fractional part, positive or negative. Real numbers also include whole numbers.
`Credited: double / float / decimal`

Character
Data type for a single letter, digit or symbol, stored using its code from the character set.
`Credited: char`

Currency
A database data type for money values, shown with a currency symbol and two decimal places.

Type check
A validation check that makes sure the data entered is of the correct data type.

Integer
Data type for whole numbers, positive or negative, with no decimal or fractional part.
`Credited: int / whole number`

Date/time
Data type that stores dates and/or times so that they can be sorted, calculated with and shown in different formats.

String / text
Data type for a sequence of characters (letters, digits and symbols) of variable length.
`Credited: text`

Boolean / Yes-No
Field type that can only hold one of two possible values, for example Yes or No (True or False).
`Credited: logical`

Exam focus

Data types come up in most Unit 1 papers, often worth one mark per item. You may be asked to match sample values to data types (2021 Q4(c), 2025 Q5(a)), choose types for database fields (2019 Q3(b), 2024 Q3(a)) or justify a choice (2019 Q3(c)). In 2023 Q1 the data types question was part of a short-answer question on data. A 'state' question needs just the type name. A 'describe' question needs what the type stores plus an example, for example 'Real stores numbers with a decimal part, such as a price of 2.85'. A justification must link to the data itself. Saying 'because it is a number' will not earn the mark; 'because it is a whole number used to count pupils' will. In database questions, Access names such as Currency, Short Text or Yes/No are also credited. The most frequent lost marks come from calling a phone number an integer.

`Unit 1 2025 Q5`	`[11]`	Data types, validation checks, data processing
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q3`	`[11]`	Field names and data types, database terms, key field, query design
`Unit 1 2023 Q1`	`[5]`	Bit and byte, data types, data in context
`Unit 1 2021 Q4`	`[6]`	Bits, nibble, byte; data types matching

? Check yourself

1. State the most suitable data type for: (a) the number of pupils in a class (b) the price of a school lunch (c) a pupil's date of birth (d) a pupil's surname (e) a grade, e.g. A. [5]
2. Explain why a mobile phone number should be stored as a string and not an integer. [2]
3. Describe the difference between the integer and real data types. [2]

Answers: 1. (a) integer (1) (b) real / currency (1) (c) date/time (1) (d) string / text (1) (e) character (1). 2. A phone number starts with 0 and the leading zero would be lost if stored as an integer (1); it may contain spaces / a + sign, which integers cannot hold (1); it is never used in calculations (1). Max 2. 3. An integer is a whole number with no decimal / fractional part, e.g. 15 (1); a real number can have a decimal / fractional part, e.g. 15.75 (1).